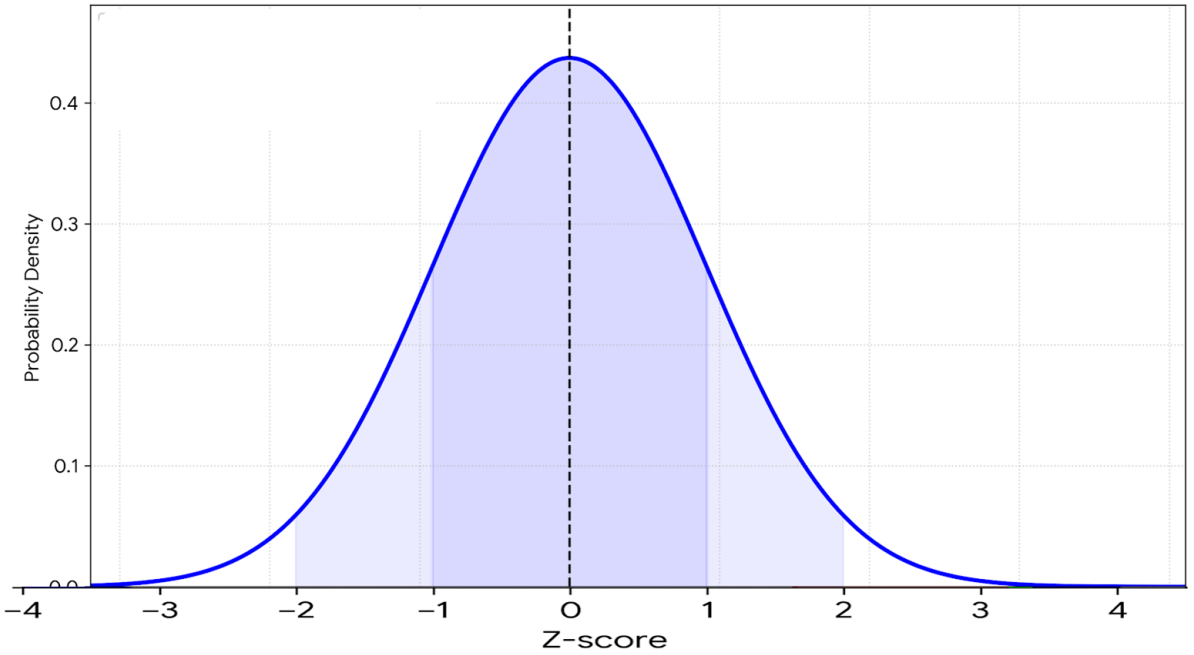
A company wants to find out if a new Machine has improved employee productivity. Before the new Machine, the average productivity was **50 units per day**, and the After the training, a random sample of **30 employees** had an average productivity of **53 units per day**. The company wants to test whether the New machine has significantly increased productivity.

Population mean:	$\mu = 50$
Sample size:	$n = 30$
Sample mean:	$x = 53$
Sample standard deviation:	$\sigma = 4.5$
CI	$= 0.97$
Significance Value	$= 0.03$

Define the Null Hypothesis
This is a **one-sample right-tailed Z-test** because:

Null Hypothesis (H_0): The New Machine program did not increase productivity.
H0 : $\mu=50$

Alternative Hypothesis (H_1): New Machine program increased productivity.
H1 : $\mu>50$



Calculate the

1) **Critical Value for Confidence level(.03) & :**
   to know the rejection Area on graph
2) **Z-score :**
   to know if the value fall under the reaction area or not
1) **Standard Error :**
   Helps to calculate the Z-score.

$$\text{Zscore } (Z) = \frac{\bar{x} - \mu}{SE}$$

$$standard\ Error\ (SE) = \frac{\sigma}{\sqrt{n}}$$

Step: Find Critical Value using t-table

Significance level α=0.03
Degree of Freedom(df)

Df =n−1
Df =30−1
Df =29
Degree of Freedom(df) = 29

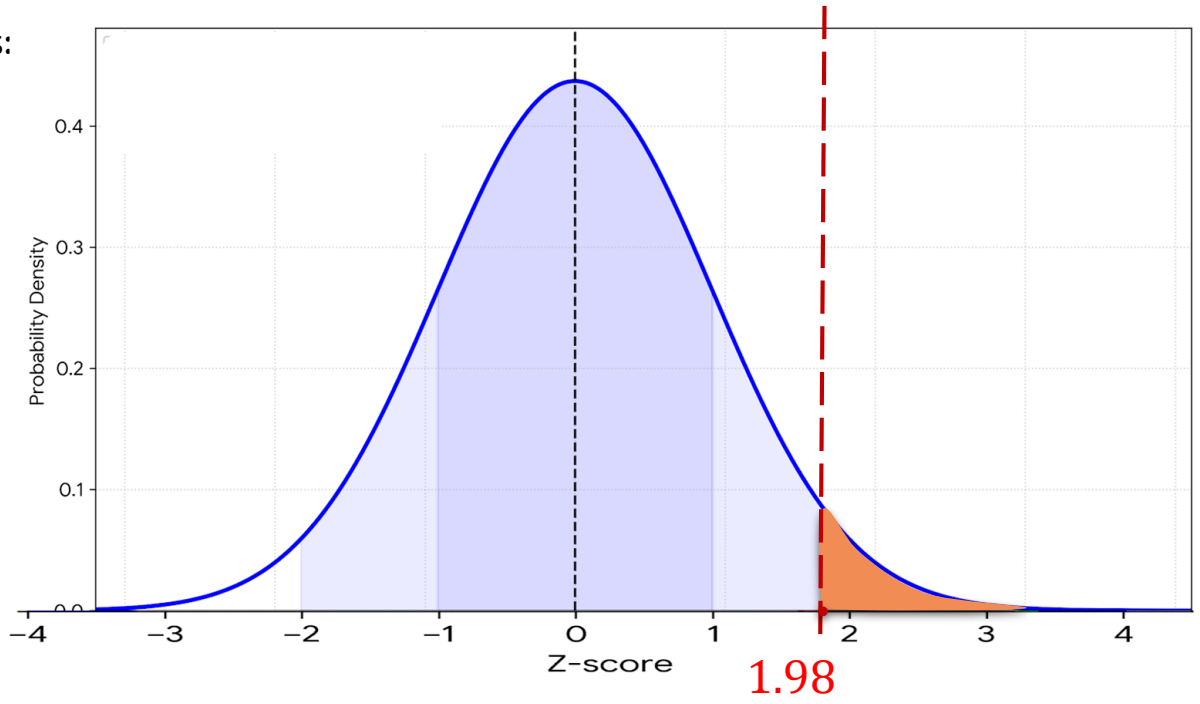
Know the critical values
to.03,df=29

Most t-tables don't directly show 0.03, so we use the closest common values:
0.050 ⟶ 1.699
0.025 ⟶ 2.045

Step A: distances in α
From 0.05 ⟶ 0.03 = 0.02
From 0.05 ⟶ 0.025 = 0.025

$$\frac{0.05-0.03}{0.05-0.025}=\frac{0.02}{0.025}=0.8$$

Difference in t:
2.045−1.699=0.346
Now take 0.8 of this gap:
0.8×0.346=0.277
Now add to lower value:
1.699+0.277=1.976
Critical Value ≈1.98



Calculate the standard error

```
Sample Standard deviation:        σ = 4.55
Sample size:                      n = 30
```

$$standard\ Error\ (SE) = \frac{\sigma}{\sqrt{n}}$$

$$standard\ Error\ (SE) = \frac{4.5}{\sqrt{30}}$$

$$standard\ Error\ (SE) = \frac{4.5}{5.5772}$$

$$standard\ Error\ (SE) = 0.8215$$

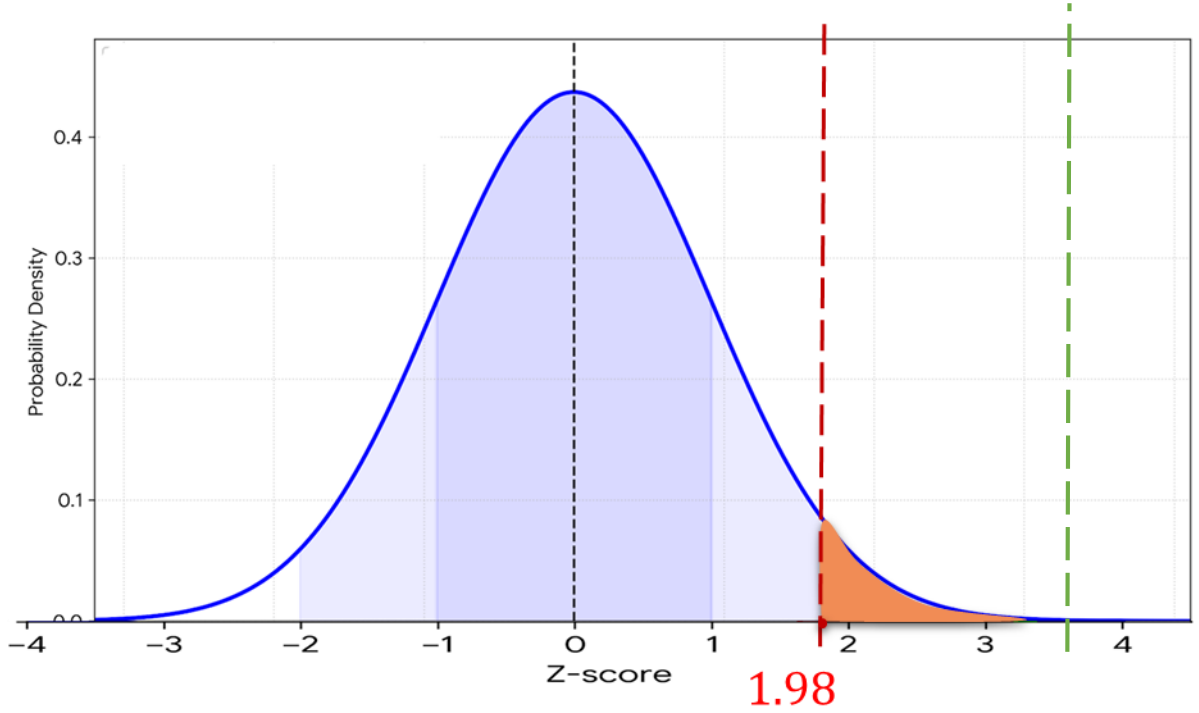
Calculate the Z-Score

```
Population mean:        μ    = 50
Standard Error          SE   = 0.8215
Sample mean:            x    = 53
```

$$\text{Zscore } (Z) = \frac{\bar{x} - \mu}{SE}$$

$$\text{Zscore } (Z) = \frac{\bar{x} - \mu}{SE} = \frac{53 - 50}{0.8216} = \frac{3}{0.8216}$$

$$\text{Zscore } (Z) = 3.65$$



Decision: Reject Ho because 3.65 > 1.98.

Conclusion: The New machine increased productivity at the alpha = 0.05 level.

Using p-Value

α	t-value
0.005	2.756
0.001	3.659

t=3.65

Since 3.65 is closer to the 3.659 we will consider 0.001

$pvalue(0.001) <$ significance level(0.03)
We can say that Null hypothesis is rejected

Conclusion
Because the p-value (0.001) is far lower than your significance level (0.03), the result is highly statistically significant.
There is a mere 0.01 chance that this productivity increase happened.
Null hypothesis is rejected.